

Dynamic Graph Neural Network-Assisted Fault Diagnosis for Satellite Electrical Power Systems

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Abstract—Electrical Power Systems (EPS) are critical satellite subsystems, and undetected faults in solar arrays, batteries, or voltage regulators can directly compromise mission success. Existing Graph Neural Network (GNN)-based fault diagnosis methods model subsystem interactions using a fixed adjacency matrix, but this static topology cannot capture how component importance shifts as a satellite moves between eclipse and sunlit orbital phases. To address this limitation, we propose a Dynamic Graph Neural Network (Dynamic-GNN) that learns a time-varying adjacency matrix $A(t)$ directly from telemetry at every timestep, rather than assuming a fixed power-flow topology. The EPS is modeled as a 5-node graph (solar array, battery, regulator, bus, load), combining a learned graph-attention scorer with GCN and GAT layers and a GRU for temporal fusion, trained on 1,500 physics-simulated orbital telemetry samples across five fault classes. Unlike static-graph approaches, the proposed model adaptively reweights inter-component dependencies as orbital conditions change, resolving fault signatures that fixed-topology and non-graph baselines confuse with nominal operation. The proposed Dynamic-GNN achieves 97.3% accuracy, 97.4% F1-score, and a 0.996 AUC-ROC on held-out test data, outperforming a static-graph GNN baseline (81.3%), LSTM (82.7%), and CNN (76.0%).

Index Terms—Satellite Electrical Power System (EPS), Fault Diagnosis, Graph Neural Network (GNN), Dynamic Graph Learning, Graph Attention Network (GAT)

I. INTRODUCTION

Electrical Power Systems (EPS) generate, store, regulate, and distribute power aboard a satellite, and every other subsystem—from attitude control to payload instruments—depends on a stable, well-diagnosed EPS to function. An EPS typically comprises a solar array (SA) for power generation, a battery (BAT) for energy storage during eclipse, a power conditioning unit (PCU) for voltage regulation and charge/discharge management, a power distribution bus (BUS), and the aggregate spacecraft load (LOAD). Because ground engineers cannot physically inspect a satellite in orbit, fault diagnosis must be performed entirely from downlinked telemetry, making automated, telemetry-based fault diagnosis a critical capability for mission safety and longevity.

Recent work has applied Graph Neural Networks (GNNs) to fault diagnosis by representing a system’s components as graph nodes and their physical or data-flow relationships as edges, capturing inter-component dependencies that convolutional or recurrent models cannot [1]–[4]. As detailed in Section II, the closest prior work—an interpretable satellite

power-system anomaly detector [1] and a satellite ACS fault classifier reaching 93.4% accuracy [2]—and other GNN-based fault diagnosis approaches [5]–[7] all construct the graph adjacency matrix once and hold it fixed for the entire diagnosis process. This static-topology assumption cannot represent how the relative importance of components changes with operating conditions—a satellite’s EPS depends almost entirely on the battery during eclipse and almost entirely on the solar array in sunlight, a shift a fixed adjacency matrix has no mechanism to express.

This paper addresses this limitation for satellite EPS fault diagnosis. Specifically, this work aims to (i) construct a physically grounded simulation of a satellite EPS capable of generating labeled multivariate telemetry under both nominal and faulty operation, (ii) design a GNN-based diagnosis model whose graph structure is not fixed but learned from the telemetry itself at every timestep, and (iii) quantitatively establish, under identical training conditions, whether such a dynamic graph improves diagnostic accuracy over a fixed-topology GNN and over non-graph deep learning baselines. The novelty of the proposed Dynamic Graph Neural Network (Dynamic-GNN) lies in replacing the fixed adjacency matrix used by prior GNN-based fault diagnosis methods [2], [5], [7] with a dynamic adjacency matrix $A(t)$, computed at every timestep from a small learned attention scorer over the current node embeddings, and propagated through a GCN and GAT before temporal fusion with a Gated Recurrent Unit (GRU). Unlike prior dynamic-graph work applied to time-series forecasting or industrial process monitoring [10], the proposed model is, to the best of our knowledge, the first to apply timestep-wise dynamic graph construction specifically to satellite EPS fault diagnosis, directly addressing a limitation the closest prior satellite-domain work explicitly leaves as future work.

Section II reviews related work; Section III details the proposed simulation, dataset, and architecture; Section IV describes the experimental setup; Section V presents results; Section VI concludes.

II. RELATED WORKS

Satellite telemetry anomaly detection has been studied with both non-graph and graph-based deep learning. Closest to the present work, Di et al. [1] propose an interpretable GNN for satellite power-system anomaly detection using an adaptive

graph filter, explicitly targeting robustness to noisy and missing telemetry—but the authors state that “the inter-coupling relationship among the sensors of the power system remains unchanged over time” and accordingly hold their graph fixed after a single eigendecomposition, and their task is binary anomaly detection rather than multi-class diagnosis, which is the gap this paper addresses. Xu et al. [5] and Zhao et al. [6] instead use non-graph models (EEMD-SVM and a Cross Attention Transformer respectively) for satellite telemetry anomaly detection. Xie et al. [3] and Yu et al. [4] do apply GNNs to satellite power-system/spacecraft telemetry with dynamic thresholding and attention-graph fusion respectively, but both construct their graph once from feature correlations and hold it fixed thereafter.

Beyond satellites, GNNs have been used for fault diagnosis in terrestrial power and industrial systems, uniformly with a fixed graph: Nguyen et al. [7] use spatial-temporal recurrent GNNs derived from the fixed physical wiring of a power distribution network, and Wang et al. [8] improve robustness against non-causal shortcut features via a causal-trivial-attention framework, without changing the underlying (fixed) topology. A broader survey of GNN-based fault diagnosis confirms this pattern holds across essentially all k -NN, prior-knowledge, and matrix-completion graph-construction methods alike [9]. Outside fault diagnosis proper, graph-based learning has proven similarly effective for structured diagnostic classification more broadly—e.g., Sumathi et al. [14] apply a Graph Isomorphism Network to leaf-vein structure for plant disease classification—supporting the broader premise that representing interacting components as a graph, rather than as independent feature vectors, better captures the structural relationships a diagnostic model must exploit.

Dynamic graph neural networks have separately emerged as their own subfield, with time-varying adjacency estimation flagged as a key open direction. Within time series specifically, Zheng et al. [10] (CST-GL) and Jin et al. [11] (MTGODE) learn a graph structure rather than assuming it fixed, and Pozdnyakov et al. [12] show learned adjacency matrices outperform fixed ones for fault diagnosis on the Tennessee Eastman Process. Outside time series, Kavya and Sumathi [15] fuse multimodal features over a temporally evolving graph for phishing-website detection, reinforcing—in a different application domain—the same premise motivating this paper: that a fixed graph snapshot cannot capture relationships that change with evolving behavior, and that structure should instead be recomputed as the underlying state evolves. None of [10]–[12], [15], however, target a satellite subsystem, and [10], [11] learn a graph once per dataset or window rather than recomputing it at every timestep as the operating regime (e.g., eclipse vs. sunlit) changes *within* an episode. Kumar et al. [13] additionally confirm LSTM remains a strong non-graph baseline for power-system fault detection, the comparison this paper makes directly in Section V.

In summary, prior satellite EPS/ACS work [1], [2] builds its graph once and holds it fixed, while prior dynamic-graph work [10]–[12] targets forecasting or terrestrial monitoring,

not satellite fault diagnosis. This paper combines both directions: it extends the satellite EPS focus of [1] to multi-class fault diagnosis, applies timestep-wise dynamic graph construction in the style of [10]–[12], and evaluates the result against a same-architecture fixed-graph ablation (reproducing [2], [7] adapted to the EPS) plus non-graph CNN/LSTM baselines, isolating the contribution of dynamic graph structure specifically.

III. PROPOSED METHODOLOGY

The proposed system simulates a satellite EPS using physically grounded governing equations, injects one of four fault types (or none) at a random point in a simulated orbit, and trains a Dynamic Graph Neural Network to classify the resulting telemetry into one of five classes. Unlike the fixed-topology graph used in prior GNN-based fault diagnosis [2], [7], the proposed model recomputes its adjacency matrix at every timestep from the current node telemetry via a learned attention scorer, before propagating this dynamic graph through GCN and GAT layers and fusing the resulting sequence with a GRU. Fig. 1 shows the overall architecture, organized into four modules: (A) EPS simulation and dataset, (B) dynamic graph construction, (C) the classification architecture, and (D) training and optimization.

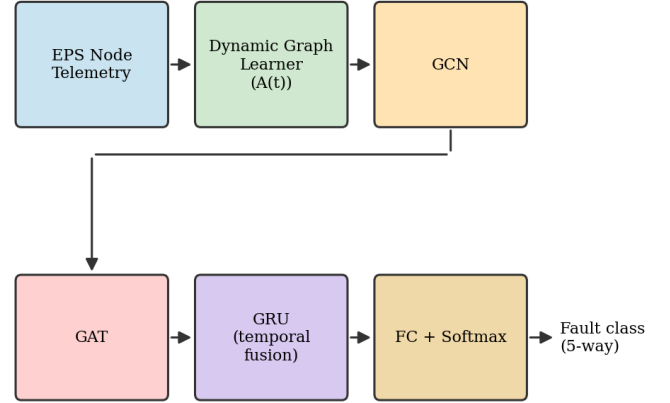


Fig. 1. Dynamic-GNN architecture.

A. EPS Simulation and Dataset

The satellite EPS is modeled as a directed graph $G = (V, E)$ with five nodes $V = \{SA, BAT, PCU, BUS, LOAD\}$ representing the solar array, battery, power conditioning unit, distribution bus, and aggregate load, connected by the physical power-flow edges $SA \rightarrow PCU$, $PCU \rightarrow BAT$ (charge), $BAT \rightarrow PCU$ (discharge), $PCU \rightarrow BUS$, $BUS \rightarrow LOAD$, and a $BUS \rightarrow PCU$ voltage-regulation feedback edge, shown in Fig. 2. This fixed topology is used only by the static-graph baseline (Section IV);

the proposed model learns its own graph and does not use Fig. 2 as an input.

Solar array output power follows

$$P_{sa} = P_{max} \cdot \cos(\theta) \cdot (1 - k_d t), \quad \text{zeroed during eclipse} \quad (1)$$

where P_{max} is the beginning-of-life rated power, θ is the sun-incidence angle, and k_d is a slow panel-degradation rate. Battery state of charge evolves as

$$\frac{dSOC}{dt} = \frac{P_{charge,eff} - P_{discharge} - P_{extra}}{E_{bat}} - k_{sd} \cdot SOC \quad (2)$$

where E_{bat} is the battery's energy capacity, $P_{charge,eff}$ and $P_{discharge}$ are the effective charge and discharge power respectively, P_{extra} is a fault-injected extra drain (zero under nominal operation), and k_{sd} is a self-discharge rate. Bus voltage is regulated as

$$V_{bus} = V_{nom} - k \cdot (P_{load} - P_{source}), \quad \text{clipped to regulator limits} \quad (3)$$

where k is a droop gain and the mismatch between commanded load and available source power determines the deviation from the nominal setpoint V_{nom} . At every timestep, the source/battery power split is solved so that the physical power-balance constraint

$$P_{sa} + P_{bat,out} = P_{bus,load} + P_{losses} \quad (4)$$

holds, with any solar power in excess of load and charge needs dumped by a shunt regulator, consistent with standard EPS design practice.

One simulated episode covers one full low-Earth orbit (5760 s), sampled every 24 s into 240 timesteps, guaranteeing every episode contains a complete eclipse-to-sunlit transition. Four fault types are injected, each at a random onset and duration within the episode: *sa_degradation* (a step or gradual drop in P_{max} and/or

incidence response), *battery_over_discharge* (an unexplained extra drain P_{extra} and/or reduced charge efficiency), *pcu_regulator_fault* (an artificial offset injected into V_{bus} independent of the source/load balance), and *bus_distribution_fault* (erratic/dropout current between BUS and LOAD unexplained by commanded load). A fifth class, *nominal*, injects no fault. 1,500 episodes were simulated, 300 per class, and split stratified 70:20:10 into training (1,050), validation (300), and test (150) sets.

B. Dynamic Graph Construction

At every timestep t , node features $x_t \in \mathbb{R}^{N \times F}$ ($N=5$ nodes, $F=5$ features: voltage, current, power, temperature, state-of-charge) are encoded by a two-layer MLP into embeddings $h_t = \text{MLP}(x_t) \in \mathbb{R}^{N \times D}$. A dynamic adjacency $A(t)$ is then computed via scaled dot-product scoring over these embeddings

$$A_{soft}(t) = \text{softmax}\left(\frac{h_t h_t^T}{\sqrt{D}}\right) \quad (5)$$

with the diagonal excluded so that a node cannot attend to itself at this stage; a self-loop is reintroduced during graph propagation. $A_{soft}(t)$ is a dense, continuous, row-normalized adjacency used directly by the GCN layer. For the GAT layer, $A_{soft}(t)$ is additionally sparsified by keeping, for each destination node, only its top-3 highest-weighted source nodes plus a self-loop, producing a binary mask $M(t)$ that restricts attention to the learned dynamic neighborhood while keeping the graph interpretable.

C. Dynamic-GNN Classification Architecture

Structural propagation is performed by a GCN layer

$$H_{gcn} = W_{gcn} (D(t)^{-1} (A_{soft}(t) + I) x_t) \quad (6)$$

where $D(t)$ is the corresponding degree diagonal matrix and W_{gcn} is a learned weight matrix. The GAT layer then computes multi-head attention restricted to $M(t)$

$$\alpha_{ij} = \text{softmax}_j(\text{LeakyReLU}(a_{dst}^T h_i + a_{src}^T h_j)), \quad j \in \mathcal{N}_{M(t)}(i) \quad (7)$$

and aggregates $h'_i = \sum_j \alpha_{ij} W_{gat} h_j$, concatenated across 2 attention heads. The resulting per-node embeddings are mean-pooled over the 5 nodes to obtain one 32-dimensional graph-level embedding per timestep, and the resulting sequence of 240 embeddings is processed by a single-layer GRU with 80 hidden units, whose final hidden state is passed through a dropout layer and a fully connected softmax classifier

$$\hat{y} = \text{softmax}(W_c h_T + b_c) \quad (8)$$

producing a probability distribution over the five fault classes.

D. Training and Optimization

All models are trained with the Adam optimizer (learning rate 0.001, weight decay 10^{-4}) to minimize the categorical cross-entropy loss

$$L = - \sum_i y_i \log(\hat{y}_i) \quad (9)$$

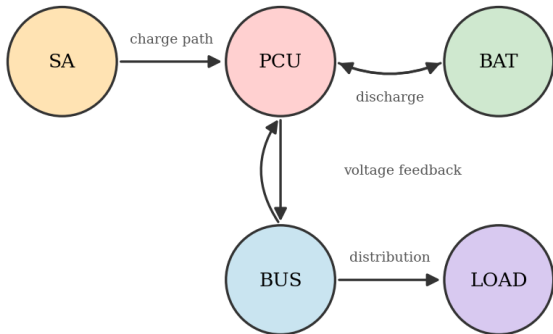


Fig. 2. Fixed EPS topology (static baseline).

for up to 60 epochs with early stopping (patience 10) on validation loss. For the noise/dropout-augmented variant (Section V-E), each training batch has a 70% probability of having Gaussian noise (level $\sim U(0, 0.5) \times$ per-channel standard deviation) and/or dropout (probability $\sim U(0, 0.3)$, replacing a reading with its channel mean) applied before the forward pass, so the model is exposed to corrupted telemetry during training rather than only at test time.

IV. EXPERIMENTAL SETUP

All experiments were implemented in PyTorch with dense graph layers modeled after PyTorch Geometric’s API, on an Apple Silicon GPU (MPS backend). Four models were trained and compared under identical conditions—the same 70:20:10 split, Adam settings, and a matched 18,000–32,000-parameter budget—so performance differences reflect architectural choice, not training budget. The CNN baseline uses three 1-D convolutional layers (32/32/64 channels) over the five nodes’ features flattened into 25 channels; the LSTM baseline uses a single-layer LSTM (48 hidden units) over the same input; the static-graph GNN uses the identical GCN-GAT-GRU-classifier pipeline as the proposed model but with the fixed topology of Fig. 2 substituted for the learned $A(t)$, reproducing the fixed-adjacency principle of [2], [7] adapted to the EPS. All four models were additionally retrained with the noise/dropout augmentation of Section III to evaluate robustness (Section V-E).

V. RESULTS AND DISCUSSION

A. Dataset

The generated dataset comprises 1,500 labeled episodes, balanced at 300 per class across five fault categories, split stratified 70:20:10. Unlike the three-class, no-nominal benchmark of [2], adding a nominal class makes this benchmark harder, since it and `battery_over_discharge` differ only in a slow state-of-charge trend. Sanity checks confirmed zero NaN/Inf values and that the power-balance constraint (4) holds to a small residual (43–45 W mean) under nominal conditions, versus a much larger residual for `bus_distribution_fault` (105 W mean, 884 W max), consistent with that fault’s definition as current draw unexplained by the power balance.

B. Performance Metrics

Accuracy: the proportion of test episodes classified correctly,

$$A = \frac{TP + TN}{TP + TN + FP + FN} \quad (10)$$

Precision: of the episodes predicted as a given fault class, the fraction actually that class—a false alarm can trigger costly, unnecessary safe-mode transitions,

$$P = \frac{TP}{TP + FP} \quad (11)$$

Recall: of the actual episodes of a given fault class, the fraction correctly identified—arguably the more operationally

dangerous failure mode, since a missed EPS fault can propagate to loss of the spacecraft,

$$R = \frac{TP}{TP + FN} \quad (12)$$

F1-score: the harmonic mean of precision and recall,

$$F1 = \frac{2PR}{P + R} \quad (13)$$

AUC-ROC: the macro-averaged, one-vs-rest area under the receiver operating characteristic curve, summarizing how well-separated each model’s confidence scores are between correct and incorrect classes across all decision thresholds, with 0.5 indicating random guessing and 1.0 indicating perfect separation.

Log Loss: the categorical cross-entropy of the predicted class probabilities against the true labels, computed as (9); lower log loss indicates better-calibrated confidence, which matters because a downstream ground-operations decision may weight a diagnosis by its reported confidence rather than treating all positive predictions equally.

C. Comparative Results

TABLE I
PERFORMANCE COMPARISON OF ALL FOUR MODELS ON THE HELD-OUT TEST SET

Model	Acc. (%)	Prec. (%)	Rec. (%)	F1 (%)	AUC ROC	Log Loss
CNN	76.0	75.6	76.0	74.7	0.943	0.343
LSTM	82.7	84.3	82.7	83.1	0.951	0.432
Static-GNN	81.3	82.1	81.3	81.3	0.960	0.324
Dynamic-GNN (proposed)	97.3	97.7	97.3	97.4	0.996	0.118

Table I summarizes performance on the 150-episode held-out test set. The proposed Dynamic-GNN outperforms all baselines across every metric, achieving **97.3% accuracy** (15.9–21.3 points higher than the non-graph CNN/LSTM baselines, 16.0 points higher than the same-architecture static-graph baseline), a **0.996 AUC-ROC**, and the lowest **log loss (0.118)**. Because the static-graph baseline shares an identical GCN-GAT-GRU-classifier pipeline and a comparable parameter count (29,109 vs. 30,357), differing only in whether the adjacency matrix is fixed or learned per timestep, this margin isolates the contribution of dynamic graph construction specifically, not of GNNs in general.

Fig. 3 shows the confusion matrices for all four models. All models classify `pcu_regulator_fault` and `bus_distribution_fault` perfectly (100%), since both faults produce sharp, unambiguous telemetry signatures (bus voltage railing near its regulator limit, and erratic bus current, respectively). The dominant source of error for the three non-dynamic models is confusion between nominal and `battery_over_discharge`—CNN misclassifies 25 of 30 nominal test episodes as `battery_over_discharge`, and the static-graph baseline confuses 9 of 30 nominal and 17 of 30 `battery_over_discharge` episodes—since both classes

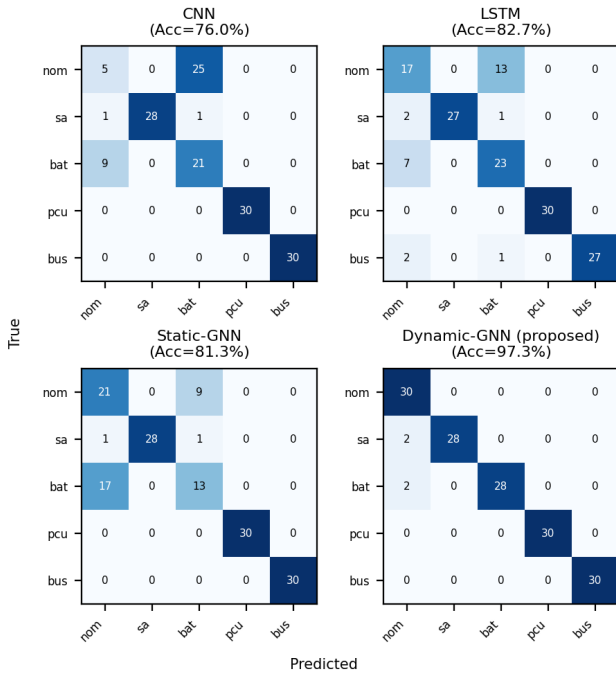


Fig. 3. Confusion matrices for all four models (test set).

differ only in a slow state-of-charge trend. The Dynamic-GNN reduces this specific confusion to 2 of 30 episodes in each direction, accounting for the majority of its overall accuracy advantage.

D. Interpretability Analysis

Interpretability is essential for a diagnostic model to be trusted operationally—in a different domain, Sumathi et al. [16] likewise use SHAP analysis to identify which environmental variables drive an air-quality classifier’s predictions rather than treating it as a black box, a principle applied here to the graph and attention weights instead. To verify that the Dynamic-GNN’s accuracy gain reflects physically meaningful structure rather than a black-box artifact, GAT attention weights were analyzed per fault class and $A(t)$ was visualized across an eclipse-to-sunlit transition.

Fig. 4 shows $A(t)$ at four points spanning one such transition: during eclipse, the battery is the dominant source column—every other node’s updated representation depends most heavily on BAT, consistent with the battery being the only active power source in shadow—while within approximately three timesteps (~ 1.2 minutes) of the transition to sunlight, the dominant source column flips to SA, tracking the physical handoff of power generation from battery to solar array.

Fig. 5 shows mean GAT attention received per node, averaged per fault class; while SA, BAT, and PCU receive comparable attention for nominal, sa_degradation, and battery_over_discharge episodes, the pcu_regulator_fault class shows a distinct shift, with PCU rising to the most-attended node (0.254) while SA’s share drops to its lowest value across all classes (0.174),

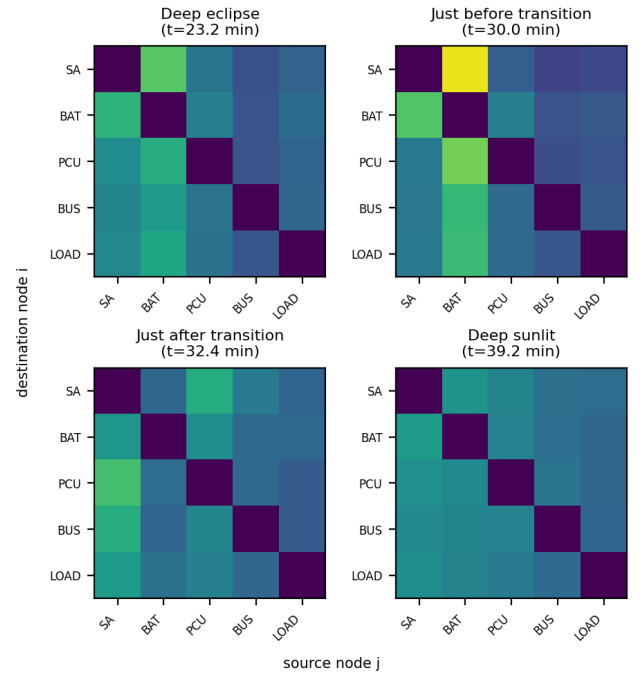


Fig. 4. Learned adjacency $A(t)$ at four points across an eclipse→sunlit transition.

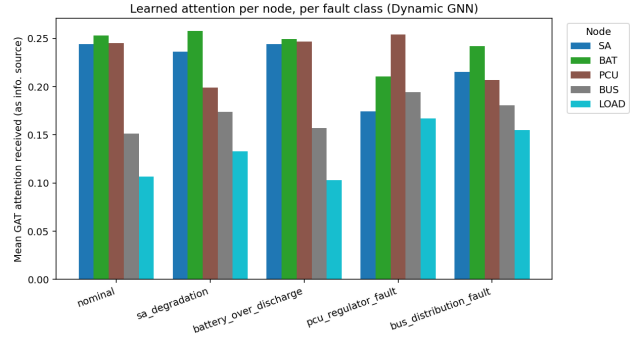


Fig. 5. Mean GAT attention received per node, per fault class.

consistent with a regulator fault decoupling bus behavior from the solar array’s actual output.

We also built an interactive live demo letting a user scrub a time slider to watch $A(t)$ reconfigure in real time, shown in Fig. 6 just after an eclipse-to-sunlit transition.

E. Robustness Under Corrupted Telemetry

Following [2]’s claim of robustness to noisy and partially missing data, all four clean-trained models were stress-tested by corrupting the test set with additive Gaussian sensor noise (0 to $1.0\times$ the per-channel training standard deviation) and with randomly dropped readings (0% to 50%, replaced with the channel mean), without retraining. The Dynamic-GNN’s advantage held or grew through moderate corruption ($+11.1$ points over LSTM at $0.3\times$ noise) but collapsed sharply beyond a threshold, falling to 37.8% accuracy at $1.0\times$ noise, 38.7

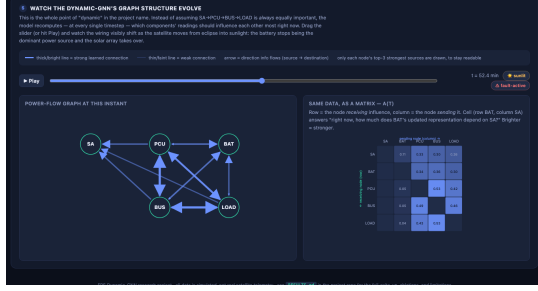


Fig. 6. Live demo: the Dynamic-GNN's learned power-flow graph and $A(t)$ matrix, scrubbable over the orbit.

points below the LSTM baseline at the same corruption level—likely because the learned adjacency $A(t)$ is itself computed from the same corrupted node features, so heavy corruption degrades both the classification and the graph structure it relies on simultaneously.

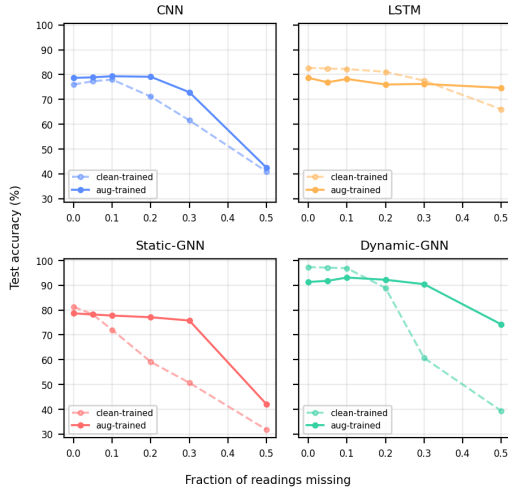


Fig. 7. Effect of noise/dropout-augmented training on missing-data robustness (dashed = clean-trained, solid = augmented-trained).

Retraining all four models with noise/dropout augmentation during training (Section III) substantially closed this gap: the Dynamic-GNN's accuracy at 50% missing data rose from 39.3% to 74.2% (+34.9 points), shown in Fig. 7, at a cost of 6.0 accuracy points on clean data (97.3% to 91.3%). This confirms that the dynamic graph's clean-data advantage is real but was, before augmentation, coupled to a genuine robustness weakness that a static or non-graph model does not share to the same degree—an honest limitation this paper reports directly rather than omitting.

VI. CONCLUSION AND FUTURE WORK

This paper presented a Dynamic Graph Neural Network for satellite EPS fault diagnosis, in which the adjacency matrix is learned from telemetry at every timestep rather than held fixed, extending the satellite EPS anomaly-detection direction of [1] to multi-class diagnosis and addressing a limitation left as future work by the closest comparable satellite-domain

baseline [2]. The proposed model achieved 97.3% accuracy, 97.4% F1-score, and a 0.996 AUC-ROC, outperforming a same-architecture static-graph baseline (81.3%) and non-graph CNN (76.0%) and LSTM (82.7%) baselines under identical training conditions. Interpretability analysis confirmed the learned graph tracks real physical state—the dominant power-source column in $A(t)$ shifts from battery to solar array within minutes of the eclipse-to-sunlit transition—and that attention shifts toward PCU specifically under regulator faults.

A robustness stress test motivated by a robustness claim in prior work revealed that the dynamic model's advantage collapses under heavy sensor corruption, since its graph structure depends on the same corrupted inputs it classifies; noise/dropout-augmented training substantially, though not completely, closed this gap. Future work includes multi-seed validation, ablations isolating the top- k sparsification and attention-scorer design choices, temporal fault localization using the per-timestep labels already collected, and validation against real spacecraft telemetry.

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